GLo-MAPPO: A Multi-Agent Proximal Policy Optimization for Energy Efficiency in UAV-Assisted LoRa Networks

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Introduction

The rapid proliferation of Internet of Things (IoT) devices presents significant challenges in energy-efficient (EE) connectivity, particularly in large-scale and non-line-of-sight (NLoS) scenarios. Long Range (LoRa) technology has emerged as a highly energy-efficient IoT communication solution [1]. As shown in Fig 1(a), the traditional terrestrial LoRa deployments face limitations such as poor coverage, inefficient resource allocation, and degraded signal reliability, especially in urban and remote environments. While the satellite-based solutions shown in Fig 1(b) extend coverage, they suffer from high latency and energy inefficiency. Recent research explores agile aerial gateways [2] using unmanned aerial vehicles (UAVs) to dynamically optimize connectivity, overcome terrestrial limitations, and enhance overall network performance. However, efficiently associating IoT devices with UAV gateways and optimally allocating network resources remain unresolved issues. Traditional optimization methods lack adaptability in dynamic multi-agent scenarios, prompting interest in deep reinforcement learning (DRL) [3] as a promising solution capable of intelligent, adaptive, and decentralized resource allocation in UAV-assisted LoRa networks.

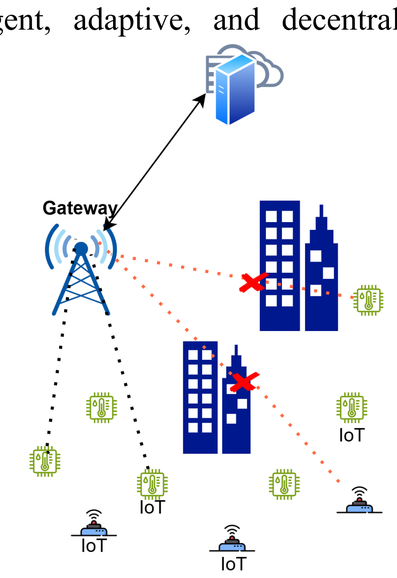

Figure 1a: Terrestrial LoRa Network deployment

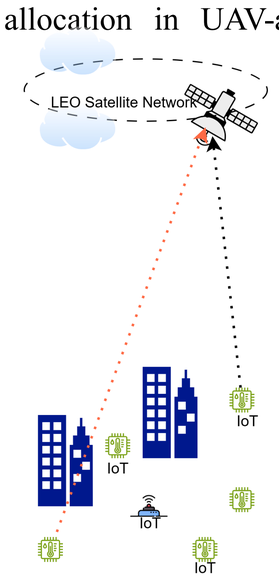

Figure 1b: Non-Terrestrial LoRa Network deployment

System Model

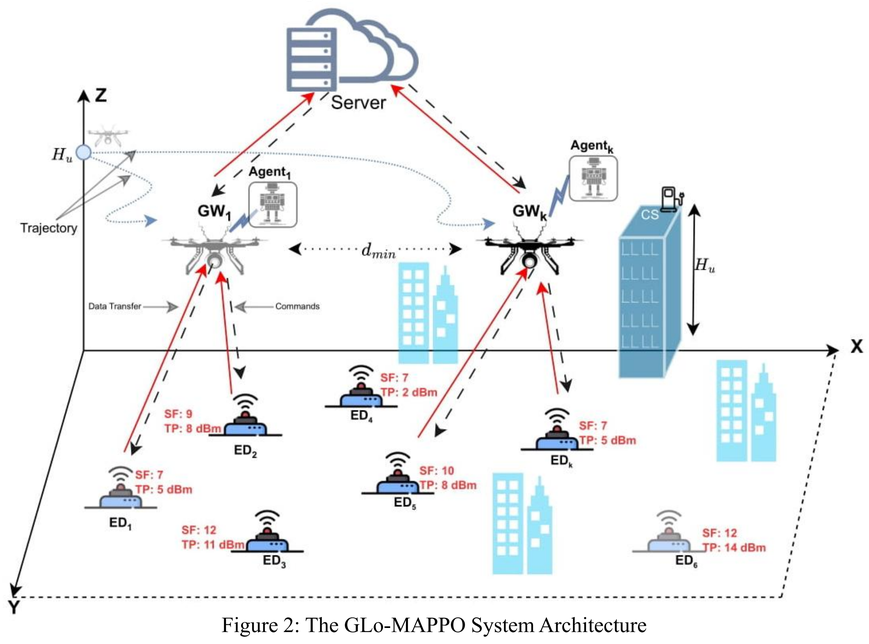

Figure 2: The GLo-MAPPO System Architecture

Problem Formulation

We aim to maximize the system's weighted global energy efficiency of flying LoRa GWs by jointly optimize **UAVs trajectories** which is determined by the UAV's movement speed and movement direction, the LoRa **spreading factor**, **transmission power** which directly affect the achievable data rates of the EDs, and **ED associations**. Consequently, our optimization problem is mathematically formulated as follows:

$$(\mathbf{P1}) \max_{\mathbf{Q},\bar{\mathbf{\Psi}},\bar{\mathbf{P}},a} \sum_{t=1}^{T}\sum_{u=1}^{U}\left[\frac{\sum_{v=1}^{V}\Re_{u,v}[t]a_{u,v}[t]}{\sum_{v=1}^{V}P_v[t]a_{u,v}[t]+P_c}\right]\cdot\lambda_u, \quad (1a)$$

$$\text{s.t.} \quad 0 \le p_{v,j}[t] \le p_{max}, \forall v \in \mathcal{V}, \forall t \in \mathcal{T}, \quad (1b)$$

$$a_{u,v}[t] \in \{0,1\}, \forall u \in \mathcal{U}, \forall t \in \mathcal{T} \quad (1c)$$

$$\sum_{u\in\mathcal{U}} a_{u,v}[t] \le 1, \forall v \in \mathcal{V}, \forall t \in \mathcal{T}, \quad (1d)$$

$$\sum_{v\in\mathcal{V}} a_{u,v}[t] \le \Lambda_{max}, \quad \forall u \in \mathcal{U},\ \forall t \in \mathcal{T}, \quad (1e)$$

$$\psi_v[t] \in \mathbf{\Psi}, \forall v \in \mathcal{V}, \forall t \in \mathcal{T}, \quad (1f)$$

$$\rho_{u,v} \ge \bar{\upsilon}_{SF}, \forall v \in \mathcal{V}, \forall t \in \mathcal{T}, \quad (1g)$$

$$x_u[t+1] = x_u[t] + s_u\cos(\theta_{ref}[t])\delta t, \quad (1h)$$

$$y_u[t+1] = y_u[t] + s_u\cos(\theta_{ref}[t])\delta t, \quad (1i)$$

$$0 \le x_u[t] \le x^{\max}, \quad \forall u \in \mathcal{U},\ \forall t \in \mathcal{T}, \quad (1j)$$

$$0 \le y_u[t] \le y^{\max}, \quad \forall u \in \mathcal{U},\ \forall t \in \mathcal{T}, \quad (1k)$$

$$\|\dot{\mathbf{q}}_u[t]\| \le S_{\max}, \quad \forall t \in \mathcal{T} \quad (1l)$$

$$d_{u,u'}[t] \ge D_{\text{safe}}, \forall u,u' \in \mathcal{U},\ u \ne u',\ \forall t \in \mathcal{T}, \quad (1m)$$

$$P_{\text{prop}}(\dot{\mathbf{q}}_u[t]) \le P_{\max}, \quad \forall u \in \mathcal{U}, \forall t \in \mathcal{T}. \quad (1n)$$

Given the complexity of our objective function, the optimization problem in Eq. (1) is challenging due to its **non-convex** and **mixed-integer nature**, making it NP-hard and difficult to solve directly.

Solution Design

Given the multiple objectives of our work, we reformulate the original problem **(P1)** to:

$$(\boldsymbol{P2}) \frac{max}{\boldsymbol{Q},\bar{\boldsymbol{\Psi}},\bar{\boldsymbol{P}},\boldsymbol{a}} \omega_1 a_u[t] + \omega_2 EE_{sys}[t] - \omega_3 D_{safe,u}[t] - \omega_4 d_{cs,u}[t]$$

subject to: constrains from (1a) to (1n).

Then, we leverage **MAPPO-based CTDE** framework, transform the optimization problem into a **POSG,** and propose **GLo-MAPPO**. The elements are described as follows:

- ***Observation Space*** *($\mathcal{O}$)* :

$$o_u[t] = \left\{\boldsymbol{q}_u[t], \kappa_u[t], \boldsymbol{q}_{cs}[t], \{\boldsymbol{g}_v[t], \boldsymbol{G}_{u,v}[t] | \forall_v \in \mathcal{V}_u[t]\}, \{\boldsymbol{q}_{u'}[t] | \mathcal{V}_{u'} \in \mathcal{N}_u[t]\}\right\}.$$

- ***Action Space*** *($\mathcal{A}$):*

$a_u[t] = \{\theta_u[t], s_u[t], \psi[t], p[t]\}$ --------> *contains both continuous and discrete.*

$a_u[t] = \{\bar{\theta}_u[t], \bar{s}_u[t], \psi[t], p[t]\}$ --------> *fully discrete.*

- ***Reward function*** *($\mathcal{R}$)*:

$$r_u[t] = \omega_1 \hat{a}_u[t] + \omega_2 EE_{sys}[t] - \omega_3 D_{safe,u}[t] - \omega_4 d_{cs,u}[t] - \boldsymbol{I}_{[t=T]}(\omega_5(1-\Xi_u^{cs}))$$

Simulation Result

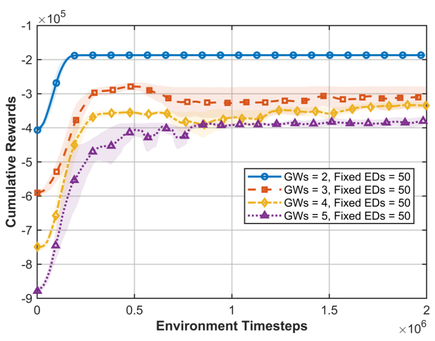

Figure 3: Varying number of flying gateways.

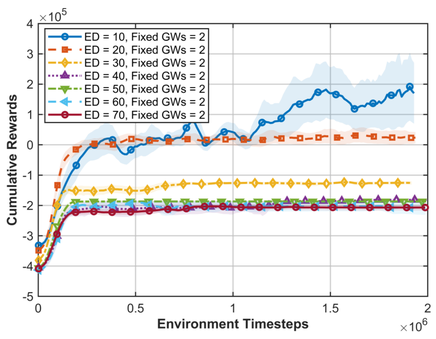

Figure 4: Varying number of LoRa end devices.

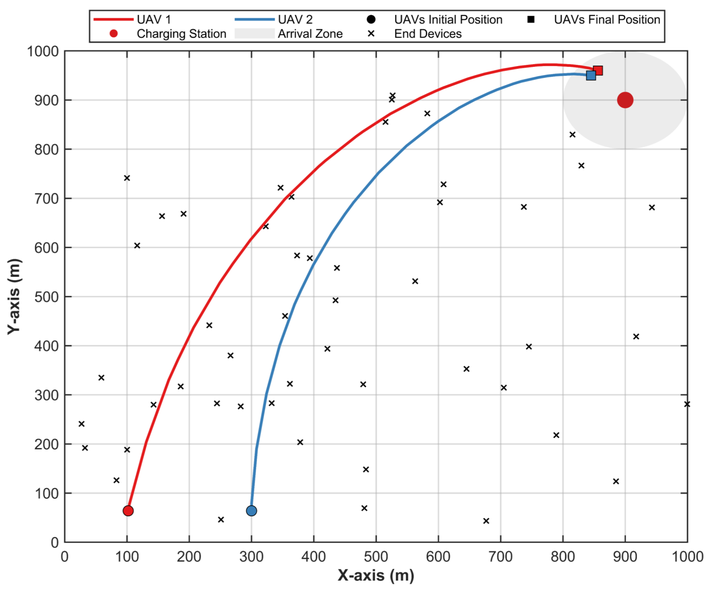

Figure 5: 2D UAV trajectory

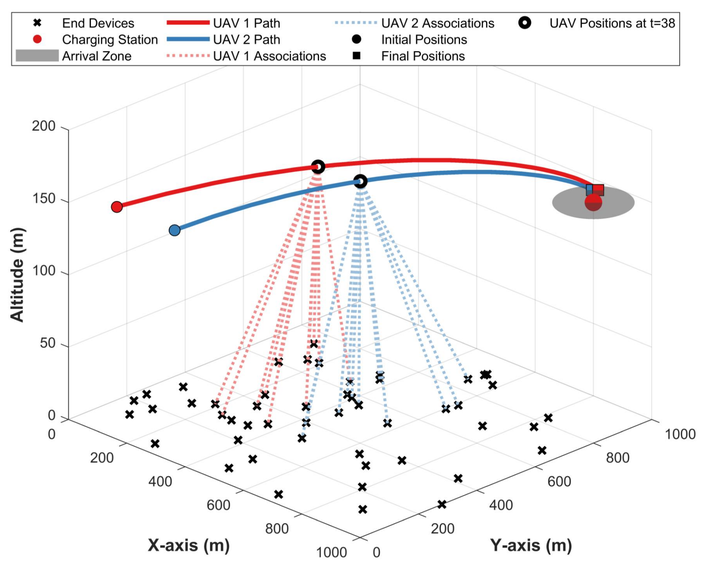

Figure 6: 3D UAV trajectory, association with LoRa end devices

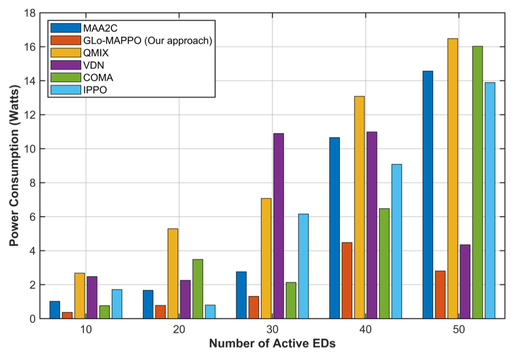

Figure 7: Power consumption with different Active LoRa end devices.

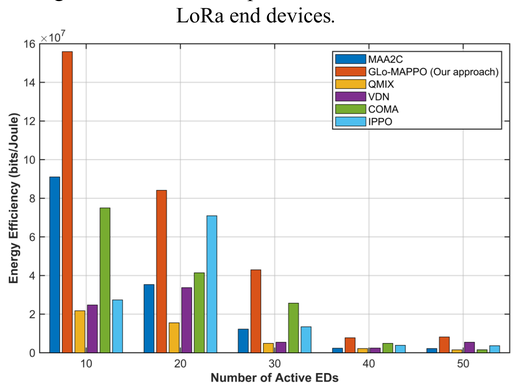

Figure 8: Energy Efficiency with different Active LoRa end devices.

Conclusion

In conclusion, we aim to optimize a system weighted global energy efficiency in an uplink data collection scenario for multi-UAV-assisted LoRa networks by formulating a multi-objective optimization problem. To archive this, we jointly optimize UAV trajectory to arrive at a given charging station, spreading factor selection, power allocation and ED association. GLo-MAPPO achieves energy efficiency improvements of **71.25%**, **18.56%**, **67.00%**, **59.73%**, and **49.95%** for EDs **10**, **20**, **30**, **40** and **50**, respectively.

References

[1] M. Jouhari, N. Saeed, M. -S. Alouini and E. M. Amhoud, "A Survey on Scalable LoRaWAN for Massive IoT: Recent Advances, Potentials, and Challenges," *IEEE Communications Surveys and Tutorials*, vol. 25, no. 3, pp. 1841-1876, 2023.

[2] Z. Zhang, S. Atapattu, B. Ren, Y. Wang, and M. Di Renzo, "Optimizing energy-efficient cooperative mac strategies for data collection in iot networks with terrestrial and non-terrestrial relays," *IEEE Internet of Things Journal*, vol. 12, no. 17, pp. 35 556–35 576, 2025.

[3] C. Yu, A. Velu, E. Vinitsky, J. Gao, Y. Wang, A. Bayen, and Y. Wu, "The surprising effectiveness of PPO in cooperative multi-agent games," in *Thirty-sixth Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2022.